12 PROJECT MANAGEMENT

CLIENT RELATIONS

1. If a surveyor agrees to locate a particular property line for a client and the client assures the surveyor that an approximate location will be adequate, which of the following statements characterizes the surveyor's best course of action?

(A) The surveyor should be certain the client understands that the monuments he establishes should not be relied upon to locate any improvements.

(B) The surveyor should word the certification of his plat in such a way as to eliminate any liability for reliance on the location he establishes.

(C) The surveyor should be mindful that people will rely on his monuments and either locate the boundary to the best of his ability or not accept the job.

(D) The surveyor should mark the line well inside his client's property to allow for a margin of safety.

Solution: The surveyor must be aware that the public will accept the correctness of his monuments without question. Location of a boundary should be undertaken with the intention of providing its correct position, or else it should not be attempted at all.

Answer is (C)

2. Which of the following statements concerning fees is not correct?

(A) The liability incurred by a surveyor is proportional to the fee that the client is charged.

(B) In determining the fee to be charged, it is proper to consider whether the work is for a regular client.

(C) In determining the fee to be charged, it is proper to consider the amount of liability involved.

(D) In determining the fee to be charged, it is proper to consider the usual charges for similar work.

Solution: Potential liability may be considered in fixing the fee; however, the liability will not be proportional to the fee charged.

Answer is (A)

3. Which of the following statements describes a service that a client cannot reasonably expect from a surveyor?

(A) A surveyor will avoid provoking disputes with the client's adjoiners.

(B) A surveyor will maintain the confidentiality of communications with the client.

(C) A surveyor will not agree to perform services for which he is not qualified.

(D) A surveyor will hold the interests of the client paramount if they come into conflict with the public interest.

Solution: The surveyor is bound to hold the public interest above the interests of the client if the two come into conflict.

Answer is (D)

4. If a regular client of one surveyor asks another surveyor to review a property survey that is currently being done by the first surveyor, which of the following questions should the potential reviewer be certain to ask?

(A) What does the client suspect is wrong with the survey?
(B) What was the fee charged by the other surveyor?
(C) How much is the client prepared to pay?
(D) Is the other surveyor aware that the client is requesting such a review?

Solution: The first consideration should be whether the other surveyor has been informed of the request for a review. In order to avoid impropriety, such a review should be undertaken only if the other surveyor has been informed and has given his consent.

Answer is (D)

5. A regular client has brought to a surveyor's office a plat made by another surveyor. The client wants his surveyor to recertify the plat. He is anxious to save money and wishes to have the recertification without incurring the cost of any more field work. What should the surveyor do?

(A) charge a fee for the recertification that is large enough to compensate him for any unforeseen liability
(B) call the surveyor who did the work and ask if he may examine the field notes from the project before recertifying the plat
(C) recertify the plat for a fee commensurate with the time and effort required to do so
(D) cordially explain to the client the reasons for refusing to do as he asks

Solution: The recertification of a plat requires that the surveyor accept full responsibility for all the information shown on the document. Clearly, a prudent surveyor is not likely to do such a thing without fully examining the entire project. A client may well be unaware of the burden assumed by a surveyor recertifying a plat. The surveyor should carefully explain his refusal to recertify the plat.

Answer is (D)

6. What is a lien?

(A) a contract describing services agreed upon between two surveyors
(B) a court order allowing a surveyor the right of entry onto private property for the performance of a survey without the owner's consent
(C) a defamation of another expressed in writing
(D) an encumbrance placed on property to secure payment

Solution: Most state laws allow a surveyor to place a mechanic's lien on the property of a client who refuses to pay for services rendered. The lien creates a nonpossessory claim against the property as security until the debt is satisfied.

Answer is (D)

7. What is the standard traditionally adopted by the courts to determine what constitutes negligence by a land surveyor when providing service to a client?

(A) whether a surveyor exercised the degree of care that a surveyor of ordinary skill and prudence would exercise under similar circumstances
(B) whether a surveyor made errors in judgment that resulted in damage to others
(C) whether a surveyor's work was reviewed and found to fall short of absolute correctness
(D) whether a surveyor exercised the degree of care that the client expected

Solution: The general standard, reinforced by many court cases, has been wheth er the surveyor performed the work with the same skill and care that is exercised by others in the profession.

Answer is (A)

8. A client and his neighbor wish to execute a property line agreement concerning their common boundary. The client has asked a surveyor to help them execute the agreement. What should the surveyor do?

(A) The surveyor should be sure that there is uncertainty on the part of both parties concerning the exact location of the property line.

(B) The surveyor should reduce the terms of the agreement to writing, describing the line and the monuments by which it may be determined, to avoid any misunderstanding as to the line agreed upon.

(C) The surveyor should first lay out the line described in the deed and show both parties the results of the measurements before they agree on any other line.

(D) Both A and B are appropriate.

Solution: A property line agreement in writing is sometimes the best solution for two adjoining property owners who are uncertain as to the location of their common boundary. However, a surveyor who insists on laying out the line described in the deed prior to such an agreement may destroy the force of the agreement itself. The courts have consistently held that the parties may decide the location of their common boundary as long as they both are uncertain of its original location and are both satisfied and clear about the line they settle upon.

Answer is (D)

9. Which of the following methods may work to discharge a contract written expressly for surveying services?

(A) mutual agreement of the parties involved

(B) impossibility of performance of the contracted work

(C) the bankruptcy of one of the parties to the contract

(D) all of the above

Solution: A contract may be discharged by performing the work to the satisfaction of the parties involved, by exercising the right to terminate the contract (where the contract expressly confers such a right), and by several other methods. All of the methods listed will discharge a contract, including the impossibility of performing the contracted work. For example, a violent storm that washes away a parcel of coastal land would terminate a contract to survey that land.

Answer is (D)

10. Which of the following statements describes a disadvantage inherent in a fixed-price contract for surveying services?

(A) Errors or ambiguities in the project's plans that become apparent after the contract has been executed can lead to disagreements.

(B) The contract may seriously impair the incentive to work efficiently and economically since every increase in cost benefits the surveyor.

(C) The actual cost of performing the work may be substantially more than the amount bid.

(D) Both A and C are true.

Solution: Usually awarded on the basis of competitive bidding, a fixed-price contract has the benefit of a high degree of certainty in cost for the client. However, there are drawbacks for the contractor, including extra work that may arise from uncertainties in plans and specifications.

Answer is (D)

ETHICS AND LIABILITY

11. Which of the following activities is not prohibited by most codes of ethics for surveyors?

(A) advertising in a self-laudatory manner

(B) practicing in an area of the profession where the surveyor is not proficient

(C) accepting remuneration other than for services rendered

(D) expressing an opinion as to who owns the land when testifying as an expert witness in court

Solution: A surveyor may express an opinion when testifying as an expert witness; in some situations, the court has allowed such opinions to involve questions of ownership. Most codes of ethics for surveyors do not prohibit such testimony.

Answer is (D)

12. In retracing the work of another surveyor, a surveyor discovers what appears to be a significant error in the position of a particular monumented corner that controls two property lines. The land involved is quite valuable and the client has recently constructed improvements in reliance on the corner. What should be the first action the surveyor takes?

(A) The surveyor should file charges against the other surveyor with the state board of registration.
(B) The surveyor should inform the client that the improvements he has constructed encroach on the adjoining property.
(C) The surveyor should check his own work.
(D) The surveyor should call the other surveyor and share his findings.

Solution: The first action a surveyor should take when finding what appears to be an error in the work of others is to thoroughly check his own work. If this does not resolve the discrepancy, the next step should be to consult with the other surveyor involved. It is quite possible that the first surveyor has information that the second is not aware of.

Answer is (C)

13. Why should a surveyor maintain a comprehensive filing system that includes all the work he performs?

(A) It is mandated by nearly all state laws that regulate land surveying.
(B) It is a professional responsibility.
(C) It is an essential strategy in limiting the surveyor's liability.
(D) It can be an excellent source of income.

Solution: An all-inclusive filing system is not required by law and will not limit the extent of a surveyor's liability for the work performed, but it is a professional responsibility. For example, a client should be able to return to the surveyor who began his project some time ago and find complete records of the work that was performed.

Answer is (B)

14. A developer is about to begin a large subdivision and has asked a surveyor to bid on the surveying necessary for the first phase. During a subsequent meeting, the developer mentions that he has received a lower estimate from one of the surveyor's competitors. What should the surveyor do?

(A) offer the developer a lower price than the competitor
(B) remind the developer that he will get what he pays for
(C) suggest a joint venture between the surveyor and the competitor on this project
(D) none of the above

Solution: Professions such as surveying do not compete on the basis of price. A client who tries to obtain an advantage by pitting surveyors against each other on the basis of their fees should find them unwilling to play along.

Answer is (D)

15. A surveyor with special expertise in riparian boundaries is asked to testify as an expert witness at a trial in St. Louis, Missouri. The offer includes the surveyor's travel expenses and a fee. The surveyor has another client in the same vicinity who wishes to consult him on a separate project. What should the surveyor do?

(A) He should inform both parties of his two objectives and divide the travel expenses between them.
(B) He should accept no remuneration for testifying.
(C) He should pay the travel expenses to St. Louis himself.
(D) There is no reason that either of the two parties need know anything of the other.

Solution: The best course is to inform both parties fully and divide the necessary travel expenses between them. It is certainly ethical to accept remuneration for expert testimony, but it is not ethical to accept any pay under what might be construed as false pretenses.

Answer is (A)

16. A contract for the surveying services necessary to construct a residential subdivision contains the following language. Which stipulation would be most likely to cause the surveyor to risk unnecessary liability or unethical practice?

(A) All surveying services will be billed per the agreed unit prices.
(B) The surveyor agrees to prepare all necessary drawings and certifications for presentation to the appropriate governmental agencies.
(C) The surveyor agrees to satisfy all state and local regulations regarding monumentation of property corners.
(D) Payment for surveying services will be rendered only after all such work has been performed to the satisfaction of the property owner and developer.

Solution: A surveyor must adhere to standards set by state regulatory agencies and its code of ethics. If these rules contradict the wishes of the client where contractual language specifies that the client must be satisfied, an irresolvable conflict may develop.

Answer is (D)

17. A surveyor is asked to perform a survey of public lands under contract to the Bureau of Land Management (BLM). However, the surveyor is not licensed in the state where the work must be done. What should the surveyor do?

(A) decline the project
(B) find a licensed surveyor in the state who will assume responsible charge for the project
(C) accept the project
(D) investigate whether the surveyor may acquire a license in the state by comity

Solution: A surveyor working in public lands exclusively under the auspices of the BLM is not required to be licensed in the state where the work is done.

Answer is (C)

ECONOMICS

18. Joe Foster is paid an hourly wage of \$18.50. He has worked 40 hr this week at the regular rate and 5.75 hr of overtime at time and a half. His net pay for the week was \$638.69. What percentage of his gross wages was deducted?

(A) 25%
(B) 30%
(C) 18%
(D) 29%

Solution: The wages may be calculated as follows.

$$\begin{aligned} (\$18.50)(40\text{ hr}) &= \$740.00 \\ (\$27.75)(5.75\text{ hr}) &= \underline{+\$159.56} \\ \text{gross wages} &= \$899.56 \end{aligned}$$

The amount received was \$638.69.

$$\begin{aligned} \text{gross wages} &= \$899.56 \\ \text{net wages} &= \underline{-\$638.69} \\ \text{amount deducted} &= \$260.87 \end{aligned}$$

$$\frac{\$260.87}{\$899.56} = 0.29$$

The deduction was 29%.

Answer is (D)

19. The Remler Surveying Company has eight employees and a monthly payroll of $35,520.00. The firm pays 11% of the gross wages into a pension plan and $35.00 per employee per month for life insurance premiums. What is the weekly cost of these fringe benefits?

(A) $3046
(B) $1996
(C) $974
(D) $2006

Solution: The monthly cost of the pension plan is

$$(\$35,520.00)(0.11) = \$3907.20$$

The monthly cost of the life insurance premium is

$$(\$35.00)(8) = \$280.00$$

Together they are

$$\$3907.20 + \$280.00 = \$4187.20$$

The weekly cost can be found by dividing the result by 4.3.

$$\frac{\$4187.20}{4.3} = \$973.77$$

Answer is (C)

20. An investment of $12,000 was made at an interest rate of 7.5% per annum. What is the value of the investment after three years?

(A) $14,907.56
(B) $14,700.00
(C) $12,930.35
(D) $14,169.20

Solution: The formula used to solve this problem is

$$P_n = P_o(1+i)^n$$

P_n is the principal at the end of the interest period, P_o is the principal at the beginning of the interest period, i is the interest, and n is the number of years of the interest period.

$$\begin{aligned} P_n &= (\$12{,}000.00)(1+0.075)^3 \\ &= (\$12{,}000.00)(1.075)^3 \\ &= (\$12{,}000.00)(1.2423) \\ &= \$14{,}907.56 \end{aligned}$$

Answer is (A)

21. A surveyor landed a lucrative government contract and needed additional equipment and personnel on short notice. He borrowed \$20,000.00. The loan was discharged five years later with a total payment of \$30,772.48. What was the interest on the loan if it was compounded annually?

(A) 9.5%
(B) 9.0%
(C) 8.5%
(D) 8.0%

Solution: Calculate the interest using the formula

$$P_n = P_o(1+i)^n$$

P_n is the principal at the end of the interest period, P_o is the principal at the beginning of the interest period, i is the interest, and n is the number of years of the interest period.

$$\begin{aligned} \$30{,}772.48 &= (\$20{,}000.00)(1+i)^5 \\ \frac{\$30{,}772.48}{\$20{,}000.00} &= (1+i)^5 \\ 1.5386 &= (1+i)^5 \\ (1.5386)^{\frac{1}{5}} &= [(1+i)^5]^{\frac{1}{5}} \\ 1.09 &= 1+i \\ 1.09-1 &= 1+i-1 \\ i &= 0.09 \end{aligned}$$

The interest was 9%.

Answer is (B)

22. A lot was purchased in January 1988 for \$10,000.00. The following end-of-year taxes were assessed: \$300.00 in 1988, \$350.00 in 1989, \$400.00 in 1990, and \$500.00 in 1991. The taxes for 1988 were paid, but the taxes for the other three years were not. At the end of 1991 the lot was sold for \$20,000.00, and the owner, now the seller, paid the back taxes at an interest rate of 9% compounded annually, as well as a 10% commission to the agent. To the nearest dollar, what portion of the \$20,000.00 selling price remained?

(A) \$15,382
(B) \$14,771
(C) \$16,648
(D) \$12,495

Solution: The formula for compound interest is

$$P_n = P_o(1+i)^n$$

P_n is the principal at the end of the interest period, P_o is the principal at the beginning of the interest period, i is the interest, and n is the number of years of the interest period.

In this problem, the taxes for 1989 and 1990 must be paid with interest. The taxes for 1988 were paid at the time, and the taxes for the current year have no interest.

1989 taxes (with interest):

$$\begin{aligned} P_n &= (\$350.00)(1+0.09)^2 \\ &= (\$350.00)(1.09)^2 \\ &= (\$350.00)(1.188) \\ &= \$415.83 \end{aligned}$$

1990 taxes (with interest):

$$\begin{aligned} P_n &= (\$400.00)(1+0.09)^1 \\ &= (\$400.00)(1.09)^1 \\ &= (\$400.00)(1.09) \\ &= \$436.00 \end{aligned}$$

1991 taxes (no interest):

$$\$500.00$$

Payment to the agent:

$$(\$20{,}000.00)(0.10) = \$2000.00$$

$$\begin{aligned} \text{selling price} &= \$20{,}000.00 \\ \text{1989 taxes with interest} &= -\$415.83 \\ \text{1990 taxes with interest} &= -\$436.00 \\ \text{1991 taxes} &= -\$500.00 \\ \text{commission to agent} &= \underline{-\$2000.00} \\ & \ \ \$16{,}648.17 \end{aligned}$$

Answer is (C)

23. What is the meaning of the term *depreciation* in financial matters?

(A) the amortizing of debt
(B) the quarterly compounding of interest
(C) the process of increasing the value of equipment
(D) the decrease in value caused by the deterioration of an asset over time

Solution: The value of all physical assets tends to decrease with use, changes in economic conditions, and technological advances. This deterioration is known as *depreciation*.

Answer is (D)

24. What is the meaning of the term *sinking fund* in financial matters?

(A) A sinking fund is an interest-bearing account into which periodic deposits are made.
(B) A sinking fund is synonymous with a depletion allowance.
(C) Establishing a sinking fund is a method of postponing a balloon payment.
(D) A sinking fund involves uniformly increasing cash flow with a uniform gradient factor.

Solution: When a firm anticipates a need for a particular sum of money at a future date for purchase of expensive equipment, it may establish a sinking fund. Regular deposits made to this interest-bearing fund are calculated to yield the particular sum needed on a particular date. A sinking fund represents a special kind of annuity.

Answer is (A)

25. An EDM purchased for \$14,000 is expected to provide service for nine years and has an estimated salvage value of \$3200. What is the annual depreciation for the instrument using the straight-line method?

(A) \$1200
(B) \$1350
(C) \$1500
(D) \$1431

Solution: Under the straight-line, or fixed-percentage, method, depreciation (the wearing value of the asset) is distributed evenly across its life according to the formula

$$D = \frac{C_o - L}{n}$$

D is the annual depreciation, C_o is the original cost, L is the salvage value, and n is the life of the asset in years.

$$\begin{aligned} D &= \frac{\$14{,}000 - \$3200}{9} \\ &= \frac{\$10{,}800}{9} \\ &= \$1200 \text{ per year} \end{aligned}$$

Answer is (A)

26. What is meant by the term *discount rate*?

(A) the rate which determines the value of an investment prior to its maturity
(B) the rate a bank is likely to charge its best customers for loans
(C) the rate of depreciation
(D) the change expected in the interest rate over a particular time period

Solution: The discount rate is used to calculate the value of an invested sum of money at some time prior to its maturation. For example, if a promissory note has a mature value of \$8000.00 and the holder wishes to know its value one year before it matures, the rate used to calculate that lesser sum of money would be the discount rate.

Answer is (A)

27. A surveyor holds a promissory note that will have a value of \$3000.00 at maturity two years hence. The surveyor has an immediate need for cash and can obtain the money by assigning the note to another party. This party will pay the surveyor today the sum that is the equivalent of the \$3000.00 two years from now. At an interest rate of 10%, what amount will be paid if the note is discounted today?

(A) \$3300.00
(B) \$2700.00
(C) \$1592.12
(D) \$2479.34

Solution: The formula for compound interest is

$$P_n = P_o(1+i)^n$$

P_n is the current value of the note, P_o is the value of the given sum of money at the date of maturity, i is the interest, and n is the number of years of the interest period. For future delivery, n is negative; the value decreases.

$$\begin{aligned} P_n &= (\$3000.00)(1+0.10)^{-2} \\ &= (\$3000.00)(1.10)^{-2} \\ &= (\$3000.00)(0.82645) \\ &= \$2479.34 \end{aligned}$$

Answer is (D)